
Structured Slot Memory for Masked-Evidence World-Model Interfaces

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Abstract

Finite-context world models can observe a decisive visual cue and later act as if it was never seen. We study this failure as an interface problem: old evidence must be stored beyond the raw context window and must also be exposed through a frozen host representation in a form a later consumer can use. We introduce a self-supervised masked-evidence JEPA objective with a compact slot-memory writer. Training never receives cue labels, cue crops, or manual key-frame annotations. Instead, salient target frames and patch sets are mined from the rendered stream, withheld from the legal context, and predicted from a causal memory state. On twelve OGBench decks, three controlled evidence ages (4, 8, and 15), and three seeds, the slot-memory method passes every registered retention cell: mean full-memory balanced accuracy is 0.983, while reset and no-state controls remain at 0.250 and 0.248. Under the same data, targets, loss, and readout, GRU, LSTM, and Mamba-lite baselines reach 0.803, 0.577, and 0.800 mean full-memory balanced accuracy and pass 22/36, 6/36, and 22/36 environment-age rows. The age-15 stress test is the clearest separator: slot memory reaches 0.962, versus 0.669, 0.350, and 0.683 for the baselines. For supported environments, fixed-controller native-use checks score 27/36 environment-age rows with 0.957 mean executed success. We scope the claim carefully: this is evidence for structured old-evidence retention and exposure under controlled ages, not autonomous planning or automatic memory-age selection.

1 Introduction

Finite-context latent world models are attractive because they compress a visual control stream into a short representation. The same finite context creates a basic ambiguity: if a cue disappears from the recent frames, is the model able to keep and expose the cue later, or did the interface silently erase it? A high-level memory claim must answer more than whether a recurrent state exists. It must show that an old observation is demanded by the task, that the evidence survives after leaving the raw context window, that it is exposed through the frozen host interface, and that a later consumer can use it.

This paper focuses on that controlled memory problem. The setting is simple by design. A rendered episode contains an early cue that determines a later target. The legal prediction context is too short to include that cue at the decision point. A memory module observes the causal stream and writes a compact state. The frozen host and audit readout are then evaluated under matched conditions: full memory, reset memory at the context boundary, and no persistent state. If full memory succeeds while the controls stay near chance, the result localizes old-evidence retention and exposure rather than a shortcut in the endpoint image or action sequence.

We instantiate this interface with a masked-evidence JEPA objective and a structured slot-memory writer. The method is intentionally non-manual. It is not told which frame is important, does not receive cue labels during training, and does not crop the cue. Instead, it mines salient frames and patch sets from the rendered stream, masks the selected target patches, and predicts fixed generic color/texture latents through a bottleneck of eight slots. Labels enter only after training through the audit readout.

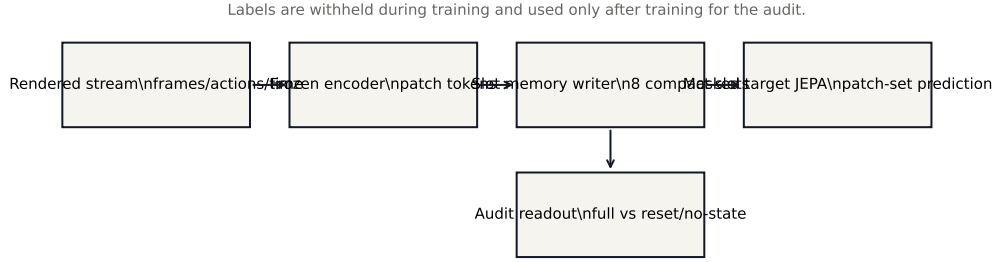


Figure 1: Masked-evidence slot-memory interface. The cue is not labeled or cropped during training. Salient target patches are mined from the rendered stream, masked from the legal context, and predicted from a causal slot memory state. Semantic labels are used only after training by the audit readout.

The main experimental question is whether the slot structure matters. We therefore compare it against three drop-in memory blocks: GRU, LSTM, and a plain PyTorch selective state-space baseline we call Mamba-lite. All four methods share the same caches, target mining, patch target encoder, loss, readout, controlled ages, and seeds. Only the memory block changes.

Contributions. First, we give a controlled old-evidence benchmark for finite-context world-model interfaces across 12 OGBench decks and evidence ages 4, 8, and 15. Second, we show that a compact slot-memory writer passes all 108 seed-level retention cells with controls at chance. Third, we add architecture baselines under the same non-manual objective and show that GRU and Mamba-lite are strong but fail many long or multi-object rows, while LSTM degrades sharply with age. Fourth, we connect readout evidence to fixed-controller native use where a local audited controller exists, while keeping the limitation explicit for AntMaze and HumanoidMaze.

2 Problem setup

Let $x_{0:T}$ be a rendered observation stream, $a_{0:T-1}$ be actions, and c be an early cue that determines a later target label y . At evaluation time, the host can see only a fixed recent window ending at time t . The cue is placed before that window, so the legal recent context cannot solve y . The memory writer receives the causal stream up to t and returns a compact state m_t . A frozen interface then exposes a representation h_t from the recent context and memory state. The audit asks whether h_t linearly exposes y .

We report three matched conditions. The full condition uses the causal memory state. The reset condition observes the same endpoint but clears memory at the context boundary. The no-state condition removes persistent memory entirely. The gate is fail-closed: full balanced accuracy must be at least 0.750 and both controls must stay at most 0.350. The exact threshold is less important than the contrast. A passing cell must show that information outside the legal context changed the exposed representation.

3 Masked-evidence slot JEPA

Figure 1 summarizes the model. A frozen frame encoder maps each rendered frame into tokens. Action and time embeddings are added to the frame token. A causal slot memory updates eight compact slots. The target side is a stop-gradient patch encoder: for each selected target patch, fixed average-pooled color, centered color, and saturation features are projected to a normalized latent. The predictor maps memory slots to a set of target latents. Training uses set InfoNCE, set cosine distance, and a small standard deviation regularizer.

Target mining without cue labels. For each episode we choose target frames from generic temporal bins spanning the legal history. Within each selected frame, high-saliency patches are mined from image statistics, not from cue metadata. These patches are removed from

Table 1: Controlled-age retention and exposure on 12 OGBench decks, three ages, and three seeds. Full, reset, and no-state are balanced accuracies from the post-hoc audit readout.

Method	Full	Reset	No-state	Seed pass	Env-age pass
Slot memory (ours)	0.983	0.250	0.248	108/108	36/36
GRU	0.803	0.252	0.250	73/108	22/36
LSTM	0.577	0.248	0.250	26/108	6/36
Mamba-lite	0.800	0.255	0.253	71/108	22/36

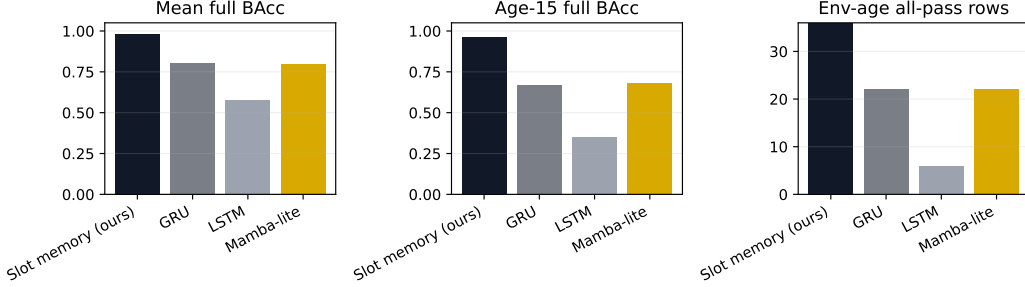


Figure 2: Architecture comparison under identical data, targets, loss, and readout. Slot memory has the highest mean readout, the highest long-age readout, and the only complete environment-age pass matrix.

the visible stream. The model must predict the fixed patch latents from the remaining causal stream. This converts the question “which old frame matters” into a self-supervised prediction pressure: the same memory state must cover old, middle, and recent visual evidence.

Slot memory. The slot writer receives the time-ordered frame/action/time tokens and a valid mask. It maintains a small set of normalized slots and predicts a set of target patch latents. The set objective prevents the model from relying on a single manually indexed memory cell. The audit then averages slot features and fits the same post-hoc classifier used for all baselines.

Baselines. GRU and LSTM baselines replace the slot writer with a recurrent sequence model. The last valid hidden state is projected into eight slots before the same target predictor. Mamba-lite is a small selective diagonal recurrence with input-dependent gates and decay. It is not the official Mamba kernel; it is a plain PyTorch state-space baseline intended to test whether a selective recurrence alone explains the gains. The baselines share all non-memory components with the slot model.

4 Experiments

We evaluate 12 OGBench decks: PointMaze-medium, PointMaze-large, PointMaze-giant, PointMaze-teleport, AntMaze-large, AntMaze-giant, HumanoidMaze-large, Cube-single, Cube-double, Cube-triple, Puzzle-3x3, and Scene. Each method is run at evidence ages 4, 8, and 15 with seeds 0, 1, and 2, giving 108 seed-level cells per method. The same render cache and target mining procedure are reused for all methods.

Table 1 is the main result. The slot-memory method passes all 108 seed-level cells and all 36 environment-age rows. The controls stay at chance. GRU and Mamba-lite are real competitors, not straw baselines: they reach 0.803 and 0.800 mean full-memory balanced accuracy, with controls near chance. But both pass only 22/36 environment-age rows. LSTM is weaker, especially as age grows, passing only 6/36 rows.

The controlled-age stress test in Table 2 is the cleanest evidence for structured memory. At age 15, slot memory reaches 0.962 mean full balanced accuracy. GRU and Mamba-lite drop to 0.669 and 0.683, while LSTM drops to 0.350. This matters because age 15 is outside the short visual context and is the closest current proxy for a long-memory requirement.

Table 2: Age-wise full-memory balanced accuracy. The age-15 row is the main long-memory stress test.

Method	Age 4	Age 8	Age 15
Slot memory (ours)	0.997	0.989	0.962
GRU	0.882	0.857	0.669
LSTM	0.775	0.605	0.350
Mamba-lite	0.904	0.812	0.683

Table 3: Fixed-controller native-use summary for supported OGBench rows. AntMaze and HumanoidMaze native-use rows are excluded because no audited local low-level controller is available.

Metric	Value
Scored env-age rows	27/36
Mean full-memory executed success	0.957
Worst scored full-memory row	0.828
Mean recent-only success	0.279
Mean random-target success	0.279
Mean oracle success	0.973

Hard rows. The failures are also informative. Puzzle and Scene are difficult for recurrent baselines because the relevant evidence is distributed across visual configuration rather than a single simple cue. GRU reaches only 0.341 on Puzzle age 15 and 0.279 on Scene age 15. Mamba-lite improves some navigation and manipulation rows but still reaches only 0.425 on Puzzle age 15 and 0.356 on Scene age 15. These rows are the main reason to prefer a structured patch-set memory objective over a generic recurrent state.

5 Native-use checks

Readout accuracy is not execution. We therefore also report a fixed-controller native-use stage where a local audited controller exists. The memory readout selects a target, and the fixed controller executes that target in the native environment. This is not autonomous planning by the world model. It is a bounded use check: if the exposed memory selects the wrong target, the fixed controller should fail.

The repaired native-use summary in Table 3 scores 27/36 environment-age rows. Full memory reaches 0.957 mean executed success, with recent-only and random target selection near 0.279. PointMaze-teleport is now included after adding a teleport-aware PointMaze controller gate. AntMaze and HumanoidMaze remain unscored for native use because this repository contains the MuJoCo environment definitions but no audited low-level expert controller.

6 Discussion

The evidence supports a focused claim. A structured slot-memory writer trained with non-manual masked-evidence targets can preserve and expose old visual evidence beyond a fixed context window. The comparison is controlled enough to attribute the main gap to the memory block: GRU, LSTM, and Mamba-lite use the same data, target mining, loss, and readout.

The evidence does not support broader claims. First, the ages are manual stress tests. The current model does not automatically infer how far back it should search; it is evaluated at prescribed delays. Second, fixed-controller native use is not native world-model planning. Third, AntMaze and HumanoidMaze native success are scope limits until an audited local low-level policy is added. Fourth, Mamba-lite is a useful selective-SSM baseline but not a claim about every possible Mamba implementation.

7 Conclusion

Controlled old-evidence memory should be evaluated as an interface property, not as a module name. Under identical masked-evidence training and readout, slot memory cleanly separates from GRU, LSTM, and Mamba-lite baselines across 12 OGBench decks and three evidence ages. The result is strong enough for a paper when stated precisely: structured slot memory improves non-manual retention and exposure beyond a fixed context window, and fixed-controller checks show that exposed memory can change native outcomes where controllers are available. Automatic memory-age selection is the next experiment, not a completed claim.

References

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A Artifact map

The paper is generated from local JSON artifacts and does not hand-enter the main numbers. The generator is `scripts/plot_paper_c.py`. It writes figures to `paper_c/figures/` and tables plus a JSON snapshot to `paper_c/generated_results/`.

Table 4: Primary local result sources used by Paper C.

Claim	Local source
Slot-memory current method	<code>outputs/multiview_patchset_color_jepa_native_v1/summary.json</code>
GRU baseline	<code>outputs/memory_arch_baselines_v1/gru/summary.json</code>
LSTM baseline	<code>outputs/memory_arch_baselines_v1/lstm/summary.json</code>
Mamba-lite baseline	<code>outputs/memory_arch_baselines_v1/mamba_lite/summary.json</code>
Fixed-controller native use	<code>outputs/native_use_repaired_combined_v2/stats.json</code>

B Training details

All methods use the same rendered caches, validation split, target mining, and readout protocol. Training uses 36 epochs, batch size 96, dimension 160, eight output slots, four attention heads for the slot method, AdamW with learning rate 3×10^{-4} , and weight decay 10^{-4} . The loss is the sum of set InfoNCE, a weighted set cosine term, and a small standard deviation regularizer. The training loss never sees the cue label. Labels are used only for post-hoc audit readouts.

C Gate semantics

A seed-level cell passes when the full-memory balanced accuracy is at least 0.750 and both reset and no-state controls are at most 0.350. An environment-age row is counted as all-pass only when all three seeds pass. We also report seed-pass counts because they show partial stability.

D Automatic age extension plan

The present paper uses controlled ages 4, 8, and 15. The next extension is an automatic age router. Instead of training separate fixed-age cells, the data loader will sample cue delays from a range and attach a timestamp to each memory slot. At inference, a query network will score stored slots by content match, timestamp, and confidence, then select the top- k slots. Evaluation will include the registered fixed ages plus unseen ages such as 6, 10, 20, and 30. We will report the maximum passing age found by a doubling and binary search protocol. This extension is intentionally not claimed by the current paper.